Pang Zixi

Singapore · e1307551@u.nus.edu

Research interests: long-term memory for LLM agents; cost–quality trade-offs in LLM systems; reinforcement learning from feedback.

EDUCATION

National University of Singapore Expected Jun 2027

Bachelor of Engineering in Computer Engineering, Year 4 *Singapore*

GPA 4.18/5.0 — Honours (Distinction) range, equivalent to Second Class Upper Honours

Selected coursework: Data Structures and Algorithms, Discrete Structures, Linear Algebra, Artificial Intelligence, Machine Learning, Software Engineering

RESEARCH EXPERIENCE

National University of Singapore — Xtra Group Jun 2026 – Present

Final Year Project, Lead Researcher · Supervisor: Prof. Bingsheng He *Singapore*

PlexusMem: Efficient Long-Term Memory for Conversational Agents

- Study the cost–quality trade-off of long-term memory for conversational agents: how memory can be built and searched with fewer LLM tokens without losing the evidence later questions need.
- Lead the project's research design: reviewed agent-memory architectures, framed its cost–quality questions, and completed the prototype design (source-linked memory units, recursive hierarchy, rule-based associations).
- Designed, ran and analyzed all experiments, including comparisons with existing memory systems, token-cost breakdowns, ablations and evidence-coverage diagnostics.

Huawei — Singapore Search & Recommendation Lab Dec 2025 – Jun 2026

Research Intern *Singapore*

PyGeoX: Reinforcement Learning for Geometric Generation

- Conducted model fine-tuning and reinforcement learning experiments for geometric generation and analyzed results.
- Worked with the first author to construct and validate PyGeoX-Bench; curated and checked Chinese entrance-examination mathematics problems for real-world evaluation.
- Wrote the manuscript's experimental section and contributed to manuscript review and revisions.

Autoregressive LLM Sampling and Interpretability

- Explored Sequential Monte Carlo decoding to reduce regeneration after late-stage rejection, comparing time–memory trade-offs with a Monte Carlo baseline.
- Investigated hidden-state-based model confidence as a signal for sample acceptance and rejection during generation.

MANUSCRIPTS & PREPRINT

Internalizing Geometric Law: Learning from Solver Residuals for Precision-Critical Generation
Rafael Cabral, **Pang Zixi**, Ziyi Shou, Shen Xin. *arXiv:2606.09278, 2026.*

Xtra-MTP: State-Efficient Parallel Verification for Hybrid Language Models
CHEN Han, Borui Xu, **Pang Zixi**, CHEN XI, Wenqi Pei, Bingsheng He, Ming Wu, Jason Zeng, Michael Heinrich.
Submitted to ICLR 2027.

ThunderWorld: Efficient World Model Generation with Selective FP4 Refinement via Native GEMMs
CHEN Han, CHEN XI, Borui Xu, **Pang Zixi**, Wenqi Pei, Feng Yu, Bingsheng He, Ming Wu, Jason Zeng, Michael Heinrich, Jiahao Yan. *Submitted to ICLR 2027.*

My role in Xtra-MTP and ThunderWorld: created manuscript figures, analyzed experimental findings, and contributed to manuscript review.

AWARD

American Express AI Hackathon — Finalist Oct – Nov 2025

Grand Finale among 50+ teams; the only undergraduate-led finalist team

- Built an AI recommendation system for business-to-merchant matching, addressing cold starts and real-time inference.

TECHNICAL SKILLS

Programming	Python (proficient), C/C++, Java, OCaml, SQL
Machine learning	PyTorch, TensorFlow, Ray/RLlib, scikit-learn, OpenCV
Tools	Git, Linux/Bash, Docker, LaTeX
Computer engineering	FPGA, embedded programming, assembly programming